

## Practice Questions (Chapter 6 - Chapter 9)

### CHAPTER 6

**Question 1 Modern Bank Runs.** Consider a bank that has assets of 100, capital of 20, and checking deposits of 80.

- 1) Set up the bank's balance sheet.
- 2) Now suppose that the perceived value of the bank's assets falls by 10. What is the bank's leverage ratio?
- 3) Suppose the deposits are insured by the government. Despite the decline in the value of bank capital, is there any immediate reason for depositors to withdraw their funds from the bank? Would your answer change if the perceived value of the bank's asset fell by 15? 20? 25? Explain.

Now consider a different sort of banks, still with assets of 100 and capital of 20, but now with short-term credit of 80 instead of checkable deposits. Short-term credit must be repaid or rolled over (borrowed again) when it comes due.

- 4) Set up this bank's balance sheet.
- 5) Again suppose that the perceived value of the bank's assets falls. If lenders are nervous about the solvency of the bank, will they be willing to continue to provide short-term credit to the bank at low interest rates?
- 6) Assuming that the bank cannot raise additional capital, how can it raise the funds necessary to repay its debt coming due? If many banks are in this position at the same time (and if banks hold similar kinds of assets), what will likely happen to the value of their assets? How will this affect the willingness of lenders to provide short-term credit?

**Question 2 Unconventional Monetary Policy.** We have written the  $IS - LM$  model in the following terms:

$$IS \text{ relation : } Y = C(Y - T) + I(Y, r + x) + G$$

$$LM \text{ relation : } r = \bar{r}$$

Interpret the real policy rate as the federal funds rate adjusted for expected inflation. Assume that the rate at which firms can borrow is much higher than the federal funds rate, equivalently that the premium,  $x$  in the  $IS$  equation is high.

- 1) Suppose that the government takes action to improve the solvency of the financial system. If the government's action is successful and banks become more willing to lend—both to one another and to nonfinancial firms—what is likely to happen to the premium? What will happen to the  $IS - LM$  diagram? Can we consider financial policy as a kind of macroeconomic policy?
- 2) Faced with a zero nominal interest rate, suppose the Fed decides to purchase securities directly to facilitate the flow of credit in the financial markets. This policy is called *quantitative easing*. If quantitative easing is successful, so that it becomes easier for financial and nonfinancial firms to obtain credit, what is likely to happen to the premium? What effect will this have on the  $IS - LM$  diagram? If quantitative easing has some effect, is it true that the Fed has no policy options to stimulate the economy when the federal funds rate is zero?
- 3) We will see later in the course that one argument for quantitative easing is that *it increases expected inflation*. Suppose it does. How does that affect  $LM$  curve?

## CHAPTER 8

**Problem 1 The Formation of Expected Inflation.** In the lecture, we have seen the following model of expected inflation:

$$\pi_t^e = (1 - \theta)\bar{\pi} + \theta\pi_{t-1}$$

- 1) Describe the process of the formation of expected inflation when  $\theta = 0$ .
- 2) Describe the process of the formation of expected inflation when  $\theta = 1$ .
- 3) How do you form your own expectation of inflation—more like 1), or more like 2)?

**Problem 2 The Macroeconomic Effects of the Indexation of Wages.** Suppose that the Phillips curve is given by

$$\pi_t - \pi_t^e = 0.1 - 2u_t$$

where

$$\pi_t^e = \pi_{t-1}$$

Suppose that inflation in year  $t + 1$  is zero. In year  $t$ , the central bank decides to keep the unemployment rate at 4% forever.

- 1) Compute the rate of inflation for years  $t$ ,  $t + 1$ ,  $t + 2$ , and  $t + 3$ .

Now suppose that half the workers have indexed labor contracts.

- 2) What is the new equation for the Phillips curve?
- 3) Based on your answer to part 2), recompute your answer to part 1).
- 4) What is the effect of wage indexation on the relation between  $\pi$  and  $u$ ?

## CHAPTER 9

**Problem 1.** The *medium-run* equilibrium is characterized by four conditions:

- Output is equal to potential output  $Y = Y_n$  and the real policy rate  $r_n$  must be chosen by the central bank so.
- The unemployment rate is equal to the natural rate  $u = u_n$ .
- The real policy interest rate is equal to the natural rate of interest  $r_n$  where  $r_n$  is defined as the policy rate where

$$Y_n = C(Y_n - T) + I(Y_n, r_n + x) + G$$

- The expected and actual rate of inflation is equal to the anchored or target rate of inflation  $\bar{\pi}$ .

- 1) If the level of expected inflation is formed so  $\pi$  equals  $\bar{\pi}$ , characterize the behavior of inflation in a medium-run equilibrium.
- 2) Write the *IS* relation as  $Y = C(Y - T) + I(Y, r + x) + G$ . Suppose  $r_n$  is 2%. If  $x$  increases from 3 to 5%, how must the central bank change  $r_n$  to maintain the existing medium-run equilibrium? Explain in words.
- 3) Suppose  $G$  increases permanently. In what direction must the central bank change  $r$  to maintain the existing medium-run equilibrium?
- 4) Suppose  $T$  decreases permanently. In what direction the central bank change  $r$  to maintain the existing medium-run equilibrium? Explain in words.
- 5) Discuss: In the medium run, a fiscal expansion leads to an increase in the natural rate of interest.